

1. Administrative & Study Identification

Official Title: A Phase I/II, Open-Label, Single-Arm, Two-Part Trial to Evaluate Safety, Tolerability, Pharmacokinetics, and Anti-Tumor Activity of Glofitamab in Monotherapy and in Combination With Chemoimmunotherapy in Pediatric and Young Adult Participants With Relapsed/Refractory Mature B-Cell Non-Hodgkin Lymphoma. **Study Title:** A Study to Evaluate Glofitamab Monotherapy and Glofitamab + Chemoimmunotherapy in Pediatric and Young Adult Participants With Relapsed/Refractory Mature B-Cell Non-Hodgkin Lymphoma **Short Title/Acronym:** iMATRIX-GLO **Protocol Number:** CO43810 **NCT Number:** NCT05533775 **Posting ID:** 981730122332301194 **Sponsor / Company Name:** Roche **Investigational Product:** Glofitamab (monotherapy and in combination with R-ICE) **Phase of Development:** PHASE1, PHASE2 **Trial Status:** RECRUITING **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics, and anti-tumor activity (specifically the Complete Response [CR] rate) of glofitamab as monotherapy and in combination with chemoimmunotherapy (R-ICE). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Complete Response (DOCR), Progression-Free Survival (PFS), the incidence and severity of adverse events, and to characterize the pharmacokinetics of glofitamab in pediatric and young adult patients.

2.2 Background & Rationale Relapsed/refractory (R/R) mature B-cell non-Hodgkin lymphoma (B-NHL) in children and adolescents/young adults is an area of high unmet medical need. T-cell engaging therapies like glofitamab (a CD20xCD3 bispecific antibody) are expected to provide substantial benefits in this population. Glofitamab redirects T cells to eliminate B cells and has already demonstrated encouraging clinical activity and a manageable safety profile in adults, prompting this evaluation in pediatric and young adult demographics.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Open-Label, Two-Part Trial split by Cohorts) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 65 participants
Number of Sites: Approximately 19 active institutions across 8 countries
Estimated Study Start: November 2022
Estimated Primary Completion: 29-Sep-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 6 months to < 18 years at the time of signing Informed Consent for Cohort A Part 1 and Cohort B of the study, and age 6 months to < 30 years old at the time of signing Informed Consent for Cohort A Part 2 of the study. 2. Histologically re-confirmed diagnosis of aggressive mature B-NHL that expresses CD20, including Burkitt Lymphoma (BL), Burkitt-like Lymphoma (BAL / mature B-cell leukemia FAB L3), DLBCL, and PMBCL. Disease must be first R/R for Cohort A and second or greater R/R for Cohort B. 3. Measurable disease (> 1.5 cm nodal, or > 1.0 cm extranodal, or bone marrow involvement) and adequate performance status (Participants < 16 years: Lansky $\geq 50\%$; Participants ≥ 16 years: Karnofsky $\geq 50\%$). 4. Adequate bone marrow, liver, and renal function. 5. Negative test results for acute or chronic hepatitis B virus (HBV), hepatitis C virus (HCV), and SARS-CoV-2 (antigen or PCR within 7 days prior). Negative HIV test at screening (stable individuals on ART with CD4 $\geq 200/\mu\text{L}$ and undetectable load may be permitted). 6. Willingness to complete clinical outcome assessments.

4.2 Exclusion Criteria 1. Isolated CNS disease of mature B-NHL without systemic involvement, or primary CNS lymphoma. Current or past history of uncontrolled non-malignant CNS disease, such as stroke, epilepsy, CNS vasculitis, or neurodegenerative disease. History of confirmed progressive multifocal leukoencephalopathy. 2. Receipt of glofitamab prior to study enrollment. 3. History of severe allergic or anaphylactic reactions to monoclonal antibody therapy or known allergy to murine products. 4. Ongoing adverse events from prior anti-cancer therapy not resolved to Grade ≤ 1 (exceptions: alopecia, Grade 2 peripheral neuropathy) or any Grade ≥ 3 adverse events. 5. Active infections not resolved prior to Day 1 of Cycle 1, or administration of a live, attenuated vaccine within 4 weeks before treatment. 6. Prior solid organ transplantation, known or suspected hemophagocytic lymphohistiocytosis (HLH), chronic active Epstein-Barr viral infection (CAEBV), or active autoimmune disease requiring treatment. 7. Major surgery or significant traumatic injury < 28 days prior to obinutuzumab pretreatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Glofitamab administered via body weight-adjusted step-up dosing on Days 8 and 15 of Cycle 1, and target dose on Day 1 of subsequent cycles. Cohort A receives glofitamab with R-ICE (rituximab, ifosfamide,

carboplatin, and **etoposide** [ATC Code: **L01CB01**; Trade Names: **Etopophos, Vepesid, Toposar**]). Cohort B receives glofitamab monotherapy. Obinutuzumab is given as pretreatment to mitigate CRS. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Multiple cycles; total duration depends on tumor response and tolerability.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care (e.g., G-CSF, anti-infectives) to manage expected cytopenias and infection risks. **Prohibited:** Other investigational agents, live attenuated vaccines, or concurrent systemic anti-cancer therapies outside the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Days -28 to 0 Informed Consent, Medical History, Tumor Assessment
Cycle 1	Days 1-10, 15-17	Inpatient Hospitalization, Pretreatment, Glofitamab Step-up Dosing, CRS Monitoring	Cycle 2
Days 1-3	Inpatient Hospitalization for target dose administration, PK sampling, Safety Labs	Cycle 3+	Day 1 onwards Continued administration, Tumor Response Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Achievement of Complete Response (CR) as determined by the investigator according to the International Pediatric NHL Response Criteria for pediatric participants and Lugano Classification for young adult participants (Arm A). 2. Percentage of participants with adverse events (AEs) (Arm A). 3. Serum concentration of glofitamab in combination with R-ICE chemoimmunotherapy (Arm A).

7.2 Secondary Endpoints 1. Objective Response Rate (ORR) (Arms A and B). 2. Duration of Complete Response (DOCR) (Arm A). 3. Progression-Free Survival (PFS) (Arm A).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Inpatient hospital monitoring during initial cycles focusing on known risks: Cytokine Release Syndrome (CRS), febrile neutropenia, tumor flare, tumor lysis syndrome, pyrexia, and infections. **Serious Adverse Events (SAEs):** Evaluated per standard protocol and CTCAE guidelines, with CRS mitigation strategies strictly employed. **Data Monitoring Committee (DMC):** Internal safety review committees ensuring the pediatric safety profile remains comparable to adult data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target \$N=65\$ participants is designed to adequately assess safety, PK parameters, and preliminary anti-tumor efficacy. **Handling of Missing Data:** Standard imputation methods for oncology trials where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite visits, source data verification, and safety reviews. **Audit Trail:** eCRF locking and data clarification forms managed via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: C043810 **Registry Number:** NCT05533775 **Posting ID:** 981730122332301194 **Sponsor/Company:** Roche **Study Title:** A Study to Evaluate Glofitamab Monotherapy and Glofitamab + Chemoimmunotherapy in Pediatric and Young Adult Participants With Relapsed/Refractory Mature B-Cell Non-Hodgkin Lymphoma.

12. Clinical Framework (Lymphoma Specific)

Primary Condition / Disease Area: Mature B-Cell Non-Hodgkin Lymphoma (Preferred UMLS Name) **Semantic Type:** Neoplastic Process **Definition:** A non-Hodgkin lymphoma that originates from mature B lymphocytes. Representative examples include diffuse large B-cell lymphoma, follicular lymphoma, marginal zone lymphoma, mantle cell lymphoma, and small lymphocytic lymphoma. **MeSH Code:** D008182 **SNOMED ID:** 127027003 **Diagnosis Threshold:** CD20-positive R/R B-NHL with bi-dimensionally measurable disease **Medical Methodology:** T-cell engaging bispecific antibody therapy +/- Chemoimmunotherapy (R-ICE) **Optimization Target:** Tumor elimination and Complete Response (CR) achievement

13. Study Procedures & Methodology

Management Pathway: Pretreatment/Step-up dosing protocol for CRS mitigation **Education Protocol (HCP):** Specialized training on Cytokine Release Syndrome (CRS) and Tumor Lysis Syndrome management **Education Protocol (Patient):** Inpatient requirement awareness and symptom tracking (e.g., fever/neurological changes)

Quality Audit Mechanism: Pharmacokinetic sampling validation and standardized tumor response tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to estimated completion (Sep 2029) **Onsite Visit Cadence:** Required inpatient stays for initial cycles (C1: Days 1-10, 15-17; C2/C3: Days 1-3) **Remote Monitoring Frequency:** Standard outpatient follow-ups per oncology protocol **Checklist Review Interval:** End of each treatment cycle for tumor assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					